Pluviago — Usage Flow

Procurement → Lab Preparation. Cultivation pipeline excluded (under development).

1. Procurement & Vendor Approval

1.1 Add an Approved Vendor · *Approved Vendor*

```
Path: Pluviago Biotech → Approved Vendor → New
```

- Open **Approved Vendor List** and click **+ New**.
- Enter the vendor's company name in **Supplier Name**. If the supplier doesn't exist yet, the system shows a confirmation popup — confirm to auto-create a new ERPNext Supplier.
- Set **Approval Status** = *Approved*, fill **Approval Date** and **Valid Upto** (the qualification expiry).
- In the **Approved Items** child table add one row per chemical the vendor is approved to supply (Item Code, Material Name auto-fills, Min Order Qty, Lead Time).
- Save. The **Vendor Code** auto-numbers as `AVL-YYYY-####`.

1.2 Raise a Purchase Order from the AVL record · *Purchase Order (ERPNext)*

- Open the Approved Vendor record.
- Click **Actions** → **Create Purchase Order**. The system creates a draft PO with the linked supplier and all approved items pre-filled with their default UOM and conversion factors.
- Edit quantities / rates as needed and submit the PO.
- If a PO is raised against a vendor whose AVL is not *Approved* or whose **Valid Upto** has expired, a soft warning fires — submission is still allowed (emergency procurement is permitted) but the warning is recorded.

1.3 Receive material — Purchase Receipt (GRN) · *Purchase Receipt (ERPNext)*

- When the chemical is delivered, create a Purchase Receipt against the PO.
- Submit the PR. ERPNext's *Quality Inspection* step is intentionally disabled for chemical items — Pluviago uses Chemical COA + Raw Material Batch instead.

2. COA Verification & Raw Material Batch

2.1 Record the vendor's COA · *Chemical COA*

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Path: Pluviago Biotech → Chemical COA → New
```

- Open **Chemical COA** → **New**.
- Pick the **Supplier** and **Item Code**. **Material Name** auto-fetches.
- Enter the vendor's **Supplier Batch No**, **COA Date**, and **Expiry Date**.
- (Optional) Click **Actions** → **Load Spec Template**. The system reads the QC Parameter Spec for this item and pre-fills the Test Parameters table with parameter name, specification, unit, and critical flag. Confirm the warning that existing rows will be replaced.
- Fill **Result Value** and **Result** (Pass/Fail) for each parameter row by reviewing the vendor's PDF COA.
- Set **Overall Result**, fill **Verified By** and **Verification Date**.
- Submit. On submit the document status becomes *Verified*; if a Raw Material Batch is linked, its **COA Verified** flag is auto-ticked.

Submit gate: the COA cannot be submitted unless Overall Result, Verified By and Verification Date are set.

2.2 Register the physical batch — Raw Material Batch (RMB) *· Raw Material Batch*

```
Path: Pluviago Biotech → Raw Material Batch → New
```

- Open **Raw Material Batch** → **New**.
- Fill **Material Name**, **Item Code**, **Supplier**, **Supplier Batch No**, **Mfg Date**, **Expiry Date**.
- Enter **Received Qty**, **UOM**, **Received Date**, and link the source **Purchase Receipt**.
- Pick **Storage Condition** (RT / 2-8°C / -20°C / 4°C) and the leaf **Warehouse** (e.g. *Chemical Store RT - PB*).
- QA reviews the COA and sets **QC Status** = *Approved*, fills **QC Checked By** and **QC Date**, then ticks **COA Verified**.
- Submit. The batch number locks (`RMB-CHEM-YYYY-####`), Remaining Qty is initialised to Received Qty, and the material becomes available for production use.

Submit gate: RMB cannot be submitted unless QC Status = Approved AND COA Verified is ticked. Expiry Date must be after Mfg Date.

2.3 Bulk-submit Drafts *· Raw Material Batch (List view)*

- From the RMB list, tick the checkboxes of one or more Draft batches.
- Click **Actions** → **Submit Selected**. Each draft is validated and submitted; the system reports how many succeeded and surfaces any failures (e.g. missing COA).

2.4 Reconcile stock if numbers drift *· Raw Material Batch*

- Open a submitted RMB.
- Click **Actions** → **Recalculate Stock**. The system re-reads every Stock Consumption Log entry for this batch and rewrites Consumed Qty and Remaining Qty.
- Use this only if the displayed remaining quantity looks wrong — Stock Consumption Log is the source of truth.

3. Stock Solution Preparation (A1 – A7)

3.1 Prepare a stock solution batch *· Stock Solution Batch*

```
Path: Pluviago Biotech → Stock Solution Batch → New
```

- Open **Stock Solution Batch** → **New**.
- Choose **Solution Type** first (A1 / A2 / A3 / A4 / A5 / A5M / A6 / A7-I … A7-VI). The batch number auto-builds as `SSB-{type}-YYYY-MM-####`.
- Fill **Prepared Date**, **Prepared By**, **Target Volume**, **Volume UOM**, **Sterilization Method**, **Storage Condition**, **Expiry Date**.
- In the **Ingredients** child table add one row per chemical: pick the source **Raw Material Batch**, the qty used, and the UOM. The system blocks any RMB that is not Approved, has no Remaining Qty, or is past expiry.
- Save the document (status: *Draft*).

3.2 Mark Preparation Complete *· Stock Solution Batch — Action button*

- Click **Actions** → **Mark Preparation Complete**.
- The system validates that at least one ingredient is present, then deducts each ingredient quantity from its Raw Material Batch via Stock Consumption Log entries.
- Status changes to *QC Pending*. The bench technician now performs the QC checks (pH, clarity, sterile filtration).

3.3 Enter QC results *· Stock Solution Batch*

- Fill **QC pH Actual**, **QC Clarity**, tick **Sterile Filtered** if applicable.
- Set **QC Status** = *Passed* and fill **QC Checked By** and **QC Date**.
- Submit the document. Status becomes *Available* and the Remaining Volume is set equal to the Actual Volume.

Submit gate: the batch must be marked Preparation Complete first, must not be Wasted, and QC Status must be Passed.

3.4 If QC fails — Mark as Wasted *· Stock Solution Batch — Action button*

- If the QC check fails, click **Actions** → **Mark as Wasted**.
- Enter the failure reason in the prompt and confirm.
- The status becomes *Wasted*. The previous stock deductions remain — they are now reflected as a loss in the Stock Consumption Log.
- Wasted batches cannot be submitted and cannot be used in any downstream medium.

3.5 See where a batch was used *· Stock Solution Batch — Action button*

- On a submitted SSB click **Actions** → **View Lineage** to see every Medium Batch that consumed this stock solution.

4. Medium Batch — Green or Red (BG-11)

4.1 Start a new Medium Batch · *Medium Batch*

```
Path: Pluviago Biotech → Medium Batch → New
```

- Open **Medium Batch** → **New**.
- Pick **Medium Type** (Green or Red) — required before saving. Naming becomes `MED-GRN-YYYY-MM-####` or `MED-RED-YYYY-MM-####`.
- Enter **Preparation Date**, **Prepared By**, **Target Final Volume**, **Storage Condition**, **Sterilization Method**, and the **DI Water RMB** with its volume.

4.2 Auto-load the recipe · *Medium Batch — Action button*

- Click **Actions** → **Load Full Formula**.
- The system reads the canonical recipe for the chosen medium type at the chosen target volume and fills both child tables — **Direct Chemicals** (e.g. CaCl■, MgSO■, NaCl for Green) and **SSB Used** (with the per-litre volumes for A1–A5 or A6 + A7-I … A7-VI).
- For each Direct Chemical row pick the source **Raw Material Batch**.
- For each SSB row pick the source **Stock Solution Batch** (must match the row's expected solution type, must be Available, must not be expired).

4.3 Mark Preparation Complete · *Medium Batch — Action button*

- Click **Actions** → **Mark Preparation Complete**.
- The system validates that direct chemicals exist and each row's RMB / SSB is valid, then writes Stock Consumption Log entries against every linked RMB and SSB.
- Status moves to *QC Pending*.

4.4 Enter the QC checkpoints · *Medium Batch*

- Green Medium: **Checkpoint 1** (clarity, pre-sterilization) and **Checkpoint 2** (pH, clarity, sterility — final).
- Red Medium: **Checkpoint 3** (clarity, pre-sterilization) and **Checkpoint 4** (pH, clarity, sterility — final).
- Fill **Sterilization Done** and **Sterilization Date**.
- Set **Overall QC Status** = *Passed* and fill **QC Checked By** + **QC Date**.
- Submit. Status becomes *Available* and Remaining Volume is set.

Submit gate: Preparation Complete must be done, batch must not be Wasted, and Overall QC Status must be Passed.

4.5 Failures and lineage · *Medium Batch — Action menu*

- If any QC checkpoint fails, click **Actions** → **Mark as Wasted**, enter a reason, confirm. Status becomes *Wasted*; stock deductions stay recorded as a loss.
- On a submitted Medium Batch click **Actions** → **View Lineage** to see every Final Medium Batch (and downstream Production Batch) that consumed this medium.

5. Final Medium Batch (75 % Green + 25 % Red)

5.1 Create from a released Green Medium · *Medium Batch (Green) — Action button*

```
Path: open a released Green Medium Batch → Actions → Create Final Medium
```

- Open a submitted Green **Medium Batch** whose status is *Available*.
- Click **Actions** → **Create Final Medium**. A dialog asks for the Target Final Volume and the Red Medium Batch to combine with.
- On submit the system creates a Final Medium Batch with Green = 0.75 × V and Red = 0.25 × V auto-calculated. The 75:25 ratio is immutable.

5.2 Validate, QC and submit · *Final Medium Batch*

- The form validates that the linked Green batch is type Green, the Red batch is type Red, and both are *Available*.
- Enter **Actual Final Volume**, **Checkpoint 5** readings (pH, clarity, sterility), and the QC sign-off fields.
- Submit. Stock Consumption Log entries deduct the used volumes from both source medium batches.

6. Stock Consumption & Traceability

- Every consumption — RMB → SSB, RMB/SSB → Medium Batch, Medium Batch → Final Medium — generates a row in **Stock Consumption Log** automatically. You do not enter these manually.
- Open **Stock Consumption Log** to audit any chemical's movement; filter by Raw Material Batch or by source DocType.
- Wasted batches still produce SCL entries (action = *Written Off (Loss)*). Cancelling a parent document writes *Reversed* entries automatically.
- Use the reports menu for end-to-end traceability:

Report	Purpose
Raw Material Traceability	Trace one RMB forward through every SSB / Medium / Final Medium / Production Batch t
Batch Genealogy	Trace one Production Batch backward through its Final Medium → Green/Red Medium →
Medium Batch Lifecycle	Status, QC outcomes and yield per Medium Batch.
SSB Consumption Report	Stock-solution usage and remaining volumes.
COA Validation History	Vendor-COA review history with verifier and decision.
QC Compliance Report	Pass / fail rates by stage.

7. Key Rules to Remember While Testing

- An RMB cannot be used until it is **Submitted** with QC Status = Approved and COA Verified ticked.
- Expired RMBs and SSBs are blocked from being added to any downstream batch.
- **Mark Preparation Complete** must be clicked before **Submit** on every Stock Solution Batch and Medium Batch — the deduction happens at that step.
- QC failure path is always: **Mark as Wasted** → **enter reason** → **confirm.** The document never ends up *Available* after that.
- Submitting a document is a hard gate — once submitted, edits use Amend (creates a new revision).
- Cancelling a submitted batch is blocked if any consumption has happened against it.
- Final Medium volumes (75 % Green + 25 % Red) are auto-calculated; do not adjust them manually.